File management:

What is a generic file type?

Generic file formats allow you to save files so that they can be opened on any platform. The files may not contain all of the formatting that can be saved in a package-specific format. Using generic formats allows files created on a PC to be read/imported on an Apple Mac and vice versa.

Common generic text files include:

- comma separated values: **.csv** file extension,
  This file type takes data in the form of tables and saves it in text format, separating data items with commas.

- text: **.txt** file extension.
  A text file is not formatted and can be opened in any word processor.

- rich text format: **.rtf** file extension.
  This is a text file type that saves some of the formatting within the text.

Common generic image files include:

- graphic interchange format: **.gif**
  This format stores still or moving images and is an efficient method of storing images using a smaller file size, particularly where there are large areas of solid colour. It is widely used in web pages.

- joint photographic expert group: **.jpg** or **.jpeg**
  This format stores still images only. It is an efficient method of storing images using a smaller file size and is widely used in web pages.

- portable document format: **.pdf**
  This is a document that has been converted into an image format. The pages can contain clickable links, video and audio. You can protect a document in pdf format to stop others from editing it.

- portable network graphics: **.png**
  It was created to replace gif and is now the most used lossless compression format used for images on the internet.

- moving picture experts group layer 4: **.mp4**
  It is a multimedia container that is used for storing video files, still images, audio files, subtitles and so on. This container is often used to transfer video files on the internet.

Common generic audio files include:

- moving picture experts group layer 3: **.mp3**
  It is a compressed file format used for storing audio files. It is suitable for use on the intrenet.

Common generic files used for website authoring include:

- cascading stylesheet: **.css**
  This is a stylesheet that is saved in cascading stylesheet format and attached to one or more web pages (often written in HTML) to define the pages' colour scheme, fonts and so on.

- hyper text markup language: **.htm** or **.html**
  This text-based language is used to create markup that a web browser will be able to display information as a web page.

Common generic compressed files include:

- Roshal archive: **.rar**
  This is a container that can hold almost any file type in a compressed format.

- Zip: **.zip**
  This is a container that can hold almost any file type in a compressed format.

<u>File compression reduces file sizes to:</u>

- save storage space
- reduce transmission time when a file is sent from one device to another.

<u>Resample an image:</u>

The process of changing the quality of an image is called resampling. Images can be downsampled meaning fewer pixels are used for the image. Images can also be upsampled by adding more pixels. Downsampling reduces the file size and therefore makes the web page load more quickly.

Headers and footers:

A header is the area of a document between the top of the page and the top margin. A footer is the area of a document between the bottom of the page and the bottom margin. You can insert text or graphics into headers and footers.

Headers and footers are needed to make sure that each page (or pair of facing pages) has elements like the page number, logo, etc. placed consistently within them. If these are placed in the header or footer, they only have to be place once but will repeat on every (or every other) page. This saves the author time and effort, not having to duplicate their work on every page.

Widows and orphans:

If you start a paragraph of text on one page or column but there is not enough room on the page to get the last line typed in, the single line of text which appears at the top of the next page or column is called a widow. Similarly, sometimes you start a paragraph at the bottom of page or column but you can only type in one line before the rest of the text goes onto the next page. The first line of the paragraph at the bottom of the page or column is called an orphan.

Breaks can be used within a document to force text onto a new page or into the next column, or to define areas with different layouts, for example where part of a document is formatted in landscape orientation and part is in portrait.

Mail merge:

A mail merged document is created to save the repeated typing of similar documents that are designed to be sent to different people. It uses a master document and a source file containing data.

Mail merge is used to save retyping or editing lots of documents. It saves time (and therefore money) and helps to reduce the number of errors that may occur in editing or retyping the document. The most common use of mail merged documents is to produce personalized letters for a number of people. The contents of the letter have parts that are the same for all people and parts that are personal to the reader.

Corporate house styles:

Most companies and organisations have a corporate house style. This is sometimes called ‘corporate branding’. This can be seen on a company’s products, printed stationery, advertising, websites and often on company vehicles.

A house style is used to make sure that all documents and other materials from an organization have consistency. It is used to save time in planning, setting up, creating and formatting documents and other materials. It is also designed to support brand recognition and reduces the risk of mistakes in documents, like typing errors in an address or telephone number, or missing important element like a logo.

Font styles:

Serif fonts are often used in newspapers and books as they are usually easier to read than sans-serif fonts. It would be appropriate to use sans-serif fonts for emphasis or for titles and subtitles. It is not sensible to use more than two different font faces on any page. You can use other enhancements to make text stand out such as bold, italics, underline and highlighting. Other elements like coloured text and backgrounds can also be used to emphasise text.

Font size:

Font sizes are measured in points. If you are asked to produce text of an appropriate size, for most adults 10 point is appropriate as body text, but older readers may prefer12 point. Anything above 14 point is generally unsuitable as body text for adults, but may be ideal for children. In stories for children learning to read (ages four to six) it may be appropriate to use 20 or 24 point font size. Larger font sizes would be appropriate

Spell Check:

Spell check is a test carried out by the word processor on the text. As you work, it checks each word and compares it to those held in its dictionary. If the words match then the word processor moves on and checks the next word. If the word does not match one in the dictionary, then it uses a red wavy underline to highlight the word to suggest it may be an error.

Sometimes words are flagged as a spelling error because the dictionary does not have a match within it. When a person's name is entered into a word processor, some names will be shown as an error and other names will not. Repeated words are flagged as spelling words.

Validation routines:

Validation is checking that data entered is reasonable. It is often a process where data is checked to see if it satisfies certain criteria when input into a computer, for example, to see if data falls within accepted boundaries.

The importance of accurate data entry:

It is critically important that data entered into computer systems is accurate.
For example: if your school stores your doctor's telephone number on its system and this number contains an error, in an emergency they may not be able to contact the doctor.
If a bank took $10000 from a bank account, rather than $10, then this would have serious consequences financially.
Errors in numeric data will cause problems if any calculations are performed.

Verification:

Verification is a way of preventing errors when data is copied from one medium to another. There are two common ways:

- **Visual verification:**

  Visual verification (visual check) is checking for data entry errors by comparing the original paper documents with the data entered into the computer.

- **Double data entry**

  Data is entered into a system twice (often by two different people). The two sets of data are then compared by the computer and any differences in the data is flagged as an error and can be corrected by the user.

Why are validation and verification needed?

Validation and verification, when used together, will help to reduce the number of errors in data entry.

You need both validation and verification because:

- Data might be sensible but has not been transcribed / transferred accurately.
- Data might have been transcribed / transferred accurately but may not be sensible.

Using validation in addition to verification would trap both errors, verification for the first example and validation for the second.

Proofreading:

Proofreading is not a form of verification. It is the careful reading and re-reading of a document (before it is finally printed) to detect any errors in spelling, grammar, punctuation or layout, whether or not they were in the original document.

Charts:

Charts are used to display series of numeric data in a graphical format to make it easier to understand large quantities of data and the relationship between different series of data.

Chart types:

Pie charts:

Pie charts are used to compare percentage values. Pie charts **compare parts of a whole** or fractions of a whole. An example would be comparing the percentage of children who preferred ice cream, jelly or trifle.

Bar charts:

Bar charts show the difference between different things. An example would be showing the number of items sold by five people in the same month.

Line graphs:

Line graphs are used to plot **trends** between two variables. An example would be plotting the temperature of water as it was heated against time.

Data manipulation:

A database is an organised collection of data. A database program is software which stores and retrieves data in a structured way. All databases store data using a system of files, records and fields:

- A **field** is a single item of data. Each field has a field name that is used to identify it within the database. Each field contains one type of data.
- A **record** is a collection of fields. These may contain different data types.
- A **file** is an organised collection of records. A file can have one or more tables within it.

There are two types of databases:

1. Flat-file databases:

A flat-file stores its data in one table, which is organized by rows and columns.

2. Relational databases:

A relational database stores data in more than one linked table, stored in a file.

Relational databases are designed so that the same data is not stored many times. Each table within a relational database will have a key field. Most tables will have a **primary key** field that holds unique data and is the field used to identify that record. Some tables will have one or more **foreign key** fields. A foreign key in one table will point to a primary key in another table. A table may have multiple foreign keys.

There are different types of relationships between fields in different tables.

- *One-to-one (1-1) relationship:*

  Each record in one table relates to only one record in another table.
- *One-to-many (1-∞) relationship:*

  One record in one table can relate to many records in another table.

Advantages of relational database over flat files:

1. Data is not repeated so less storage capacity is used.
2. Data is not repeated so each change to an item of data has to be made only once.
3. It is easier for users to produce reports from a relational database, where data is held in two or more tables, than from two or more flat-file databases.

Although people often think that it is quicker to search using relational rather than flat-file databases, it is not always the case. It depends on the structure of both databases and the quantity of the data being searched.

Data Types (field type):

- Alphanumeric/text
- Numeric:
  - integer
  - decimal
  - currency
  - date and time
- Boolean/Logical

Other data types can be found in commercial databases, for example placeholders for media such as images, sound bites and video clips. These are often used in web applications where a back end database holds the media to be displayed in another application, such as a webpage.

Searching using wildcards:

A wildcard is a character that is used as a substitute for other characters. The * (asterisk) character to show a number of characters (including 0), while the ? (question mark) to show a single character.

Key features of a well-designed form:

The most important feature of form design:

- keep the form simple, with clear questions, using closed questions where possible. This will limit the different answers to be stored in the database and will make it easier to search the database.
- similar fields grouped, but not crowded, together with white space between each data entry box.
- a title that states what data is being collected
- instructions on filling the form
- not just the field names but written questions. Example, 'What is your first name?'
- appropriate space for each field for the data that will be added and space between each field
- radio buttons (or drop down menus) are used where possible
- navigation buttons on the form to allow a user to add new records and move between records
- important data like the key field can be highlighted to show that this data must be completed before the record can be saved

Presentations:

A presentation is a series of slides used to give information to an audience.

A presentation can be used in many different ways: to teach or inform as visual aid in a lecture, or as a constant on-screen carousel giving information or advertising, for example in a shopping mall.

The media for delivery and type of presentation developed will depend on the purpose of the presentation and the target audience. Different media will require different screen/page sizes.

Most presentations will require a consistent colour scheme and consistently applied styles to all slides.

Consistency is really important in the development of your presentations; simple themes and colour schemes using one or two fonts save presentations from being messy and disorganised. A well-structured and organised presentation usually says to the audience 'I am a well-organised and reliable person'. One way of doing this is to use a master slide.

Always use the same transition effect between slides and the same animation effect throughout the whole presentation. Consistency in these areas is just as important as using consistent styles and colour schemes.

Audience and presenter notes:

Delivery of a presentation with a multimedia projector may include the use of audience notes and/or presenter notes.

Audience notes:

Audience notes are paper copies of the slides of a presentation that are given to the audience so that they can take them away and refer to them after the presentation. Sometimes people will want to write their own notes on their audience note printouts during the presentation. These can be printed in different formats, with several slides on a page, or just one slide with space for the person to add their own notes.

Presenter notes:

Presenter notes are a single copy of the slides from a presentation, with key facts that need to be told to the audience by the person delivering the presentation.

Data analysis:

Spreadsheet basics:

A spreadsheet is sometimes called a sheet or a worksheet. Many sheets can be held within a single workbook.

A worksheet or sheet is a single page in a file created with an electronic spreadsheet. A worksheet is used to store, manipulate, and display data.

A spreadsheet is a two-dimensional table split into rows and columns.

Columns run vertically in a worksheet. Each column is identified by a letter in the column header starting with Column A and running through to Column XFD.

Rows run horizontally in an Excel worksheet. Each row is identified by a number in the row header. There are more than one million rows in each Excel worksheet.

In a spreadsheet program, each rectangular box in a worksheet is referred to as a cell. Each cell has an address. A cell is the intersection point of a vertical column and a horizontal row. Data entered is stored in a cell. Each cell can hold only one piece of data at a time. The contents of a spreadsheet cell can be:

- a number
- text, which is called a label
- a formula, which always starts with an = sign.

The changing of cells to see the results is called modelling.

Tabs:

Accuracy of data entry:

One error, for example a mistyped number or decimal point in the wrong place, could cause all the data in the spreadsheet to be incorrect. Care must also be taken when entering a formula, as one small error is likely to stop the spreadsheet working as it is expected to.

Named cells and ranges:

When an individual cell or an area of a spreadsheet is going to be used a number of times within the formulae of a spreadsheet, it is often a good idea to give it a name. This name should be short and meaningful. In the case of a large spreadsheet, it is easier to remember the name of a cell rather than trying to remember the cell reference. Once a cell or a range of cells has been named, you can use this name in all your formulae.

Formulae and functions:

Formulae in spreadsheet programs are used to perform calculations on values entered and stored in the program. A function is a pre-set formula.

Mathematical operators used in formulae:

- For addition use the + symbol
- For subtraction use the – symbol
- For multiplication use the * symbol
- For division use the / symbol
- Indices are calculated using the ^ symbol.

If you combine several operators in a single formula, Excel performs the operations in the following order:

1. %
2. ^
3. * and /
4. + and -

If a formula contains operators with the same precedence — for example, if a formula contains both a multiplication and division operator — Excel evaluates the operators from left to right.

To change the order of evaluation, enclose in parentheses the part of the formula to be calculated first.

To check that the formulae are correct, compare you original paper-based calculations with the values in the spreadsheet.

Nested formulae and functions:

A nested formula or function is having one formula or function inside another one. Sometimes nested formulae could contain several formulae nested within each other.

Relative and absolute referencing:

Relative referencing: By default, a spreadsheet cell reference is *relative*. What this means is that as a formula or function is copied and pasted to other cells, the cell references in the formula or function change to reflect the function's new location.

Absolute referencing: An absolute cell reference consists of the column letter and row number used in a regular cell reference but both letter and number are preceded by dollar signs (**$**).
Examples of absolute cell references would be C4, G15, or A345.
One of the main uses for absolute cell references is in a formula when you want a cell reference stay fixed on a specific cell.
As a result, if the formula is copied and pasted to other cells, the absolute cell references in the formula or function do not change.

Test the data model:

Designing a test plan and choosing your test data are the most important parts of testing the data model. If you test every formula of the spreadsheet thoroughly the number of possible errors is reduced when you use the spreadsheet with real data. Choose data that will test every part of a condition. If you are testing calculations, use simple numbers that make it easier for you to check the calculations. Be careful to test each part of the spreadsheet with **normal** data that you would expect to work with your formulae, with **extreme** data to test the boundaries and with **abnormal** data that you would not expect to be accepted.

Carefully check that each formula and function works as you expect it to by using simple test data.

Write down each number and the expected results before trying each number. Check that the actual result matches the expected result for every entry. If not, change the formula before starting the whole test process again.
Sample test plan:

Data entry in B3	***Data type***	***Expected result***	***Actual result***
	Extreme/Normal		
	Normal		
	Abnormal	*Error – value not available*	

It is very important to check carefully ranges within formulae and functions.
Check that all formulae and functions work before using real data in your model. If you find an error during testing, correct it, and then perform all of the tests again as one change to a spreadsheet can affect lots of different cells.

Excel Functions

SUM, AVERAGE, MAX, MIN:

Function	Equivalent formula	What it does
=SUM(B4:B8).	=B4+B5+B6+B7+B8	
=SUM(B4,B8)	=B4+B8	
=SUM(B4:B6,B8)	=B4+B5+B6+B8	
=SUM(*MyRange*)		Adds up the contents of all the cells within a named range called *MyRange*
=AVERAGE(B4:B8)	=(B4+B5+B6+B7+B8)/5	
=MAX(B4:B8)		To select the largest figure within the list of numbers
=MIN(B4:B8)		To select the smallest figure within the list of numbers

INT & ROUND:

Function	Value given	What it does
=INT(62.5512)	62	Takes the whole number part of a number and ignores all the digits after the decimal point
=ROUND(62.5512,2)	62.55	Rounds to two decimal places
=ROUND(62.5512,1)	62.6	Rounds to 1 decimal place.
=ROUND(62.5512,0)	63	Rounds to 0 decimal places
=ROUND(62.5512,-1)	60	Rounds to the nearest 10.
=ROUND(62.5512,-2)	100	Rounds to the nearest 100.

COUNT, COUNTA:

COUNT is used to count only the cells with numbers in them.

COUNTA is used to count the number of non-blank cells.

To count the number of text values both COUNTA and COUNT functions are used.

IF:

If a number of nested IF statements are used be careful to work in a logical order. Work from smallest to largest or vice versa.

COUNTIF & SUMIF:

COUNTIF looks at the cells within a given range and counts the number of cells in that range that meet a given condition.

SUMIF compares each value in a range of cells and, if the value matches the given condition, it adds the value in another related cell to form a total.

LOOKUPS:

The term 'look up' means to look up from a list. There are three variations of the LOOKUP function that can be used. These are:

LOOKUP is used to look up a value using data in the first row or the first column of a range of cells and returns a relative value.

HLOOKUP is a function that performs a horizontal look up of data.

VLOOKUP is a function that performs a vertical look up of data.

LEFT, RIGHT, MID:

Function	Value given	What it does
=LEFT("AHMED",1)	A	
=LEFT("AHMED",2)	AH	
=RIGHT("AHMED",1)	D	
=RIGHT("AHMED",2)	ED	
=MID("AHMED",2,1)	H	Returns the characters from the middle of a text string, given a starting position and length
=MID("AHMED",2,2)	HM	
=MID("AHMED",3,2)	ME	

Website authoring:

Web development layers:

Website:

A website is a collection of individual but related web pages that are often stored together and hosted by a web server. A web page is created using three layers:

- The content layer (structure layer): to enter the content of a web page and create web page structure.
- The presentation layer: to format whole web page/s or individual elements.
- The behavior layer: to enter scripting language to a web page or an individual element.

You develop the content/structure layer of your web pages in a language called HTML and the presentation layer of your web pages in CSS.

HTML - HyperText Markup Language:

HTML is a text-based language used to develop the content/structure layer of websites. Files are written in HTML using a simple text editor or web-authoring package.

HTML uses a set of markup tags to describe a webpage to the web browser. The web browser does not display HTML tags but uses them to display the content of the page.

HTML tags should be in lower case.

Each webpage has two sections: the head and the body. The head section starts with <head> and close with </head> and objects between those tags are not usually displayed by the web browser. The head section should always contain a title. This is the name displayed in the browser toolbar. It is the page title used if a page is added to your ‘favorites’ in your browser and is the title displayed in search engine results.

The body section starts with <body> and close with </body> and objects between these tags will be displayed in the web page.

Comments in HTML starts with <!-- and end with -->.

Tables:

Tables are used to create the basic structure of many web pages. They are used to organise page layout and are often used in web pages even though the borders may not be visible.

Hyperlinks:

A hyperlink is a method of accessing another document or resource from your current application.

- **Hyperlinks within a web page - Anchors**

  A division is a point of reference within a web page. It is similar to a bookmark when using word-processing. If you create a web page that will not fit in a single window, it is useful to use one division for each section of the web page, so the user can move to any section without having to scroll through the whole document. An anchor is used to set a hyperlink to allow you to navigate within the page or navigate to an external page. An anchor starts with an <a> tag and closes with an tag. If the anchor name is visible in the browser view of the page it often means you have made a syntax error.

- **Hyperlinks to other web pages**

  Hyperlinks can be created to another web page stored locally, usually in the same folder as the current web page, or to an external website on the internet.

  Make sure you do not put an absolute address in a hyperlink reference as this is only likely to work on your computer. Other computers are unlikely to have the same folder structure and filename.

- **Hyperlinks to send an email message**

Relative file path and absolute file path:

Absolute paths always include the domain name of the website, including **http://www.**, whereas relative links only point to a file or a file path. When a user clicks a relative link, the browser takes them to that location on the current site. For that reason, you can only use relative links when linking to pages or files within your site, and you must use absolute links if you're linking to a location on another website.

So, when a user clicks a relative link, the browser looks for the location of the file relative to the page where the link appears.

Cascading stylesheet - CSS:

A cascading stylesheet is a simple way of adding style (for example, fonts, colours, spacing) to web pages.

One or more of these cascading stylesheets can be attached to a web page, and the styles in the stylesheet will be applied to this page. Where more than one web page is used, the styles only have to be defined once and attached to all the web pages. This makes all the pages have similar appearance. In-line styles usually over-ride styles attached from an external stylesheet.

If more than one stylesheet is attached to a web page at the same time, those attached later in the markup have priority over earlier ones. If a style has more than one declaration of the same property, the last value is used for the property.

To add a comment to a stylesheet use /* before the text and */ after the text.

CSS format:

CSS rules have a selector and a declaration block look like this.

Selector	Declaration		Declaration	
↓	↓		↓	
h1	**{color:**	**#ff0000;**	**Font-size:**	**14px;}**
↑	↑	↑	↑	↑
HTML element	Property	Value	Property	Value

When attaching an external stylesheet to a web page, make sure that you do not put an absolute address in a hyperlink reference as this is only likely to work on your computer. Other computers are unlikely to have the same folder structure.

Publish a website:

Every web page that you create should be stored in a single folder to make sure that all the page elements are kept together for uploading a website to the internet. There are many ways of creating and uploading a website and its elements to the internet. It can be hosted on your computer, but this is rarely done as few of us have the hardware and enough bandwidth on our internet connection to do this. Many people use hosting companies in order to do this and upload a website into their hosting space.

All websites have a domain name, such as www.hodder.co.uk, which is used to find the site. To publish your website you must register the domain name you wish to use. You will use FTP, which means file transfer protocol, to upload your files to your web hosting space.

Test a website:

Before testing takes place it is important to understand the purpose of the website and web page, and the target audience for the page. As much as possible every element of a website should be tested before it is uploaded to the web server.

A test plan should be developed to make sure that you do not miss anything.

Testing consists of two phases: functional testing and user testing.

<u>Functional testing:</u>

All page elements must be checked to ensure that they appear as you expected.

This will include for each web page:

- Is the table structure correct?
- Do all images appear as planned?
- Are all objects that are not supposed to be visible hidden from the user?
- Do all internal hyperlinks work?

For the entire website test:

- Can each page be found from the expected URL?
- Do all links between pages work as expected?
- Do all external hyperlinks open the correct web pages?

An example of part of a test plan for a web page:

Test	Expected outcome	Actual outcome	Remedial action
Is table stylesheet attached?	Table format as in design specs	Yes	
Is font stylesheet attached?	Fonts as in design specs	No	Edit h4 to be #ff0000 rather than f80000
Table borders hidden	Hidden	Visible	Set table attribute to border=" " Then restart
Hyperlink from **Publisher** goes to www.hodder.co.uk	Opens Hodder home page	Yes	

<u>User testing:</u>

The plan:

1. Decide what needs to be tested.
2. Find a suitable test audience between two and five users or potential users. Do not use IT specialists unless that is who the site is designed for.
3. Tell the users it is the website being tested not them, and that you value their thoughts and opinions.
4. Ask the users to speak their thoughts as they work but do not respond.
5. Observe the test so that difficulties can be noted but do not help in any way.

The test may include:

- tasks to complete using the site
- questions to answer
- navigation to the page to be tested from the home page.

Here is an example of some user test questions.

Let the user view the web page for 15 seconds then ask:

1. *What are your first impressions of the web page?*

Give the user sufficient time to read the web page content, then ask:

2. *What is the purpose of the web page?*
3. *Was this purpose clear from the beginning?*
4. *How easy is it to read and understand?*
5. *Is there too much or little information?*
6. *What did you like about the web page?*
7. *What did you dislike about the web page?*
8. *Did you experience any difficulties on the web page?*
9. *What was the overall quality of your experience?*
10. *What could be done to improve the web page content of the presentation?*
11. *Do you have any other comments or suggestions?*

<u>Justify the choice of test plan:</u>

You must be able to identify which elements of the test plan are functional testing and which are user testing. You must be able to make decisions on why and how you might test some elements of a web page. Make it clear why you have made your choices and not made other choices.

Example:

Create one element of a functional test plan and one part of a user test plan for this website. Each text item is a hyperlink to a new page and a stylesheet, **rockstyle.css**, has been used. For each element justify your choices.

Answer:

Functional test: The hyperlink from 'News' goes to the News page. I must test that this link works so that I can navigate through to this web page; if the link is broken the page will not work. The original page may have contained a reference including a drive letter and path that may not appear on another computer.

User Test: I would select a range of test users from the target audience who are aged between 14 and 40. I would select these test users because they are more likely to want to listen to rock music than young children or older people.

References:

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